# PENGZHEN LI

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## EDUCATION

**University of Tennessee, Knoxville** expected May 2027
Ph.D. in Economics

**Shandong University, Jinan** 2018
M.A. in Finance — thesis published in *Technological Forecasting and Social Change* (2019); 1,000+ citations

**Tongji University, Shanghai** 2015
B.A. in Economics

## RESEARCH INTERESTS

Environmental and Energy Economics, Agricultural and Resource Economics, Applied Econometrics, and International Trade.

## PEER-REVIEWED PUBLICATIONS

*Four SSCI/SCI-indexed publications; 1,132 total citations, h-index 4 (Google Scholar, August 2026).*

"Do green technology innovations contribute to carbon dioxide emission reduction? Empirical evidence from patent data." K. Du, **P. Li**, and Z. Yan. *Technological Forecasting and Social Change*, 146 (2019), 297–303. **1,000+ citations (Google Scholar).**

"Spatial optimization of the sustainable aviation fuel supply chains from forest residues via fast pyrolysis/hydrotreatment considering feedstock ash content variability." **P. Li**, T. E. Yu, N. Labbé, N. Abdoulmoumine, M. Garcia-Perez, K. P. Hoyt, and B. C. English. *Biomass and Bioenergy*, 208 (2026), 108793.

"Assessing the impact of preprocessing and conversion technologies on the sustainable aviation fuel supply from forest residues in the Southeast USA." **P. Li**, T. E. Yu, C. Trejo-Pech, J. A. Larson, B. C. English, and D. N. Lanning. *Transportation Research Record*, 2678(9) (2024), 639–654.

"Beef price spread relationship with processing capacity utilization." C. Martinez, **P. Li**, C. N. Boyer, T. E. Yu, and J. G. Maples. *Journal of the Agricultural and Applied Economics Association*, 2(1) (2023), 133–145.

## WORKING PAPERS

**Selling Down-Market: Market Access and the Quality of Chinese Exports [JMP].**
Abstract: Trade liberalization can change export quality through cheaper inputs or through access to foreign buyers. I study the latter, exploiting the 2001 US grant of permanent normal trade relations to China, which eliminated the threat of tariff increases that varied across products. Comparing the same firm's US and non-US sales of a product before and after 2002, I find that quality fell about 5 percent more in highly exposed products than in barely exposed ones. The fall appears on impact, does not detectably reverse, and has no counterpart in the EU. Prices fell, but quantities rose less than the price decline alone predicts. I develop a model in which access lets firms sell down-market, to buyers who value quality less.

**Trade Restrictiveness Indices with Discriminatory Tariffs** with James Lake.

Abstract: A Trade Restrictiveness Index (TRI) is the uniform tariff that is welfare-equivalent to a country's actual tariff structure. Its standard implementation assigns a single tariff to each product, yet preferential trade agreements make tariffs on the same product differ across exporters, distorting where a country buys as well as what it buys. In an Armington model, we derive a TRI that separates these two distortions: for a small country, the squared index is, to second order, the square of the conventional index evaluated at mean product tariffs plus a term in the variance of tariffs across each product's exporters. We also derive the index for large importers, whose tariffs affect world prices. Across 123 importers over 2000–2020, the dispersion term accounts for 24 percent of the median importer's squared TRI in 2018, up from zero in the early 2000s. Preferences therefore reduce the TRI far less than they reduce average tariffs: in 2018 they cut the median importer's simple average tariff by 16 percent but its TRI by only 5 percent.

**Price Risk, Irreversible Investment, and the Welfare Benchmark for the Section 45Z Clean Fuel Production Credit** with T. Edward Yu. [paper] [slides]

**How Should 45Z Pay for Risk? A Welfare-CVaR and Project-Finance Decomposition of the IRA Clean Fuel Production Credit** with T. Edward Yu. [paper] [slides]

**Estimating Pork Primal Price Relationships Using a Threshold Vector Autoregressive Analysis** with Charles Martinez, Christopher N. Boyer, T. Edward Yu, and David Anderson. *SSRN working paper 4121953 (2022).* [paper]

**Estimating Large Games with a Payoff Threshold: Candidates and Party Lists in Brazil** with Matt Van Essen and Luiz Lima. *2024 J. Fred and Wilma Holly Award (best second-year paper).* Abstract: Estimating a discrete-choice game of incomplete information requires evaluating every joint action profile, which is infeasible when dozens of candidates interact, as in Brazilian council elections. We instead adopt a mean-field equilibrium, in which candidates affect one another only through list shares, reducing the fixed point to one dimension per list. Because seats go only to lists that clear a threshold, an equilibrium exists if and only if some proposed set of clearing lists is self-fulfilling, and we show that such a set need not exist. Since several lists can clear the threshold in the same market, within-market log-odds contrasts difference out unobserved list appeal and identify how responsiveness differs across candidate types. In markets of a few dozen candidates, sampling noise in list shares attenuates the maximum-likelihood estimate of the weight on the chance of a seat; a split-sample instrument removes most of this bias in simulations. Across 926,900 council candidacies, candidates favor lists with a better chance of a seat, and among non-incumbents the response is stronger for those with a university degree.

## WORK IN PROGRESS

**A Weight-Free Benchmark for Multilateral Tariff Bargaining** with Georg Schaur. *Draft, August 2026.* [paper] [slides]

## RESEARCH EXPERIENCE

**Graduate Research Assistant**, University of Tennessee 2021 – 2023
USDA NIFA (two projects) and FAA Office of Environment and Energy (ASCENT) funded research on forest-residue biofuel and sustainable aviation fuel (SAF) supply chains in the Southeast U.S.

- Designed cost-minimizing and profit-maximizing supply-chain networks with mixed-integer programming (GAMS); results published in *Biomass and Bioenergy* and *Transportation Research Record*.
- Facility-siting and investment analysis combining econometric models with the IMPLAN regional impact platform; efficiency analysis of facilities and airports with Data Envelopment Analysis.

- Sensitivity and scenario analysis of feedstock quality and market conditions with extended mathematical programming (EMP), supervised machine learning, and @RISK.
- Managed high-resolution geospatial data in ArcGIS, R, and Python.

**Other research experience**

- *Panel and causal methods:* synthetic-control estimate of the long-run output effect of the 2008 Wenchuan earthquake; GMM estimates of MFP/CFAP payments on U.S. agricultural lending; Hansen panel-threshold models of green innovation and $CO_2$; mediation analysis of digital finance and emissions for Chinese listed firms.
- *Time series:* threshold VAR of regime switching in commodity-price linkages; VECM and multivariate GARCH of wholesale commodity-price volatility.
- *Machine learning:* drivers of $CO_2$ emissions from patent data; equity-premium prediction from *NYT* and *WSJ* text.

## PRESENTATIONS

**Western Regional Science Association**, Annual Meeting, Hawaii 2023
**Western Agricultural Economics Association**, Annual Meeting, Santa Fe, NM 2022
**Southern Agricultural Economics Association**, Annual Meeting, New Orleans, LA 2022
**Midwest Economic Theory and International Trade Meetings**, Virginia Tech Spring 2025
**University of Tennessee**, Economics Brown Bag Seminar Series Spring 2024, Fall 2024
**University of Tennessee**, Applied Micro Group Spring 2024, Fall 2024

## TEACHING EXPERIENCE

**University of Tennessee**

***Instructor***

ECON 351: Monetary Economics Spring 2025, Spring 2026
ECON 322: The Global Economy: Trade and Development Summer 2025
ECON 313: Intermediate Macroeconomics Summer 2026

***Teaching Assistant***

Ph.D. courses:
ECON 511: Microeconomic Theory Fall 2023, Fall 2024, Fall 2025
ECON 512: Advanced Microeconomics Spring 2024
ECON 581: Mathematical Methods in Economics Summer 2024
Master's courses:
AREC 525: Agribusiness Operations Research Methods Spring 2022, Spring 2023

## HONORS AND AWARDS

Charles B. Garrison Award for Excellence in Teaching (best teaching), University of Tennessee 2026
Summer Graduate Assistantship, University of Tennessee 2025
J. Fred and Wilma Holly Award (best second-year paper), University of Tennessee 2024
J. Fred and Wilma Holly Fellowship, University of Tennessee 2023 – present
Graduate Assistantship, University of Tennessee 2021 – present
Academic Scholarship for Master's Students, Shandong University 2017
Freshman Scholarship, Shandong University 2016
Outstanding Rainbow Volunteer, Tongji University (tutoring pupils in Shanghai) 2013 – 2015
National Endeavor Scholarship, Tongji University 2014
Tongji Scholarship of Excellence, Third Prize, Tongji University 2014

## SKILLS

**Programming Languages:** Stata, R, Python, SQL, GAMS, MATLAB, LaTeX, ArcGIS.
**Methods:** panel and causal econometrics (FE, IV, DiD, event study, SCM, GMM, panel threshold), time series (TVAR, VECM, mGARCH), mixed-integer and stochastic programming, DEA, cooperative and mean-field games, machine learning.
**Credentials:** CFA Level II.
**Languages:** Chinese (native), English (fluent).

## REFERENCES


**T. Edward Yu**
Professor of Agricultural and Resource Economics
Department of Agricultural and Resource Economics
University of Tennessee
Email: tyu1@tennessee.edu

**Georg Schaur**
Leigh Burch III Economics Professor and Haslam Family Faculty Fellow
Department of Economics
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**James Lake** (Chair)
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